

Qualification Pack



In-Vitro Diagnostic (IVD) Service Engineering and Maintenance Supervisor

QP Code: LFS/Q6401

Version: 1.0

NSQF Level: 5.5

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Qualification Pack

Contents

LFS/Q6401: In-Vitro Diagnostic (IVD) Service Engineering and Maintenance Supervisor	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	4
LFS/N6401: Introduction to IVD, Biomedical Instrumentations and Physiological Measurements	5
LFS/N6402: Bio-Signal Acquisition and Safety in Biomedical Instrumentation	8
LFS/N6403: Analytical Principles & Instruments Used for Clinical Chemistry	11
LFS/N6404: Operations of Biomedical Transducers and Sensors in IVD	14
LFS/N6405: Internet of Things and Smart Systems for IVDs and Medical Devices	17
LFS/N6406: Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics ...	20
LFS/N6407: Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs	23
LFS/N6408: Operations and Servicing of Point of Care IVDs	26
LFS/N6409: Perform installation, routine maintenance and trouble shooting following the guidelines for IVD component assembly and material handling guidelines.	29
LFS/N6410: Manage Customer Satisfaction through field service and maintenance operations	32
DGT/VSQ/N0102: Employability Skills (60 Hours)	35
Assessment Guidelines and Weightage	42
<i>Assessment Guidelines</i>	42
<i>Assessment Weightage</i>	43
Acronyms	45
Glossary	46



Qualification Pack

LFS/Q6401: In-Vitro Diagnostic (IVD) Service Engineering and Maintenance Supervisor

Brief Job Description

This short-term, practical-oriented course will enable learners to understand the fundamentals of medical diagnostics, the operating principles of In-Vitro Diagnostic (IVD) equipment, and its applications, to take up Service Engineer/Technician jobs in the IVD segment of the Medical Devices Sub Sector. To equip learners with industry-relevant skills in mechanical systems, manufacturing processes, automation, regulatory practices, biomedical engineering concepts, and medical electronics, which are crucial for the efficient servicing and maintenance professional to follow guidelines set by production, quality assurance, and field application of In Vitro Diagnostics (IVD).

Personal Attributes

A Service Engineer and maintenance individual in the IVD segment should possess strong analytical and problem-solving abilities, attention to detail, and a solid technical aptitude to handle biomedical and electronic equipment effectively. The individual should demonstrate accountability, adaptability, and a willingness to continuously learn emerging technologies and regulatory practices. Good communication skills and a customer-centric approach are essential for interacting with healthcare professionals, while teamwork and time management support efficient service delivery. Additionally, the person must maintain high ethical standards, ensure compliance with safety and quality guidelines, and exhibit resilience while working in high-pressure healthcare environments.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LFS/N6401: Introduction to IVD, Biomedical Instrumentations and Physiological Measurements](#)
2. [LFS/N6402: Bio-Signal Acquisition and Safety in Biomedical Instrumentation](#)
3. [LFS/N6403: Analytical Principles & Instruments Used for Clinical Chemistry](#)
4. [LFS/N6404: Operations of Biomedical Transducers and Sensors in IVD](#)
5. [LFS/N6405: Internet of Things and Smart Systems for IVDs and Medical Devices](#)
6. [LFS/N6406: Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics](#)
7. [LFS/N6407: Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs](#)
8. [LFS/N6408: Operations and Servicing of Point of Care IVDs](#)
9. [LFS/N6409: Perform installation, routine maintenance and trouble shooting following the guidelines](#)



Qualification Pack

[for IVD component assembly and material handling guidelines.](#)

10. [LFS/N6410: Manage Customer Satisfaction through field service and maintenance operations](#)

11. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
Country	India
NSQF Level	5.5
Credits	24
Aligned to NCO/ISCO/ISIC Code	NA
Minimum Educational Qualification & Experience	B.Sc (Medical Lab technology, Biochemistry, Chemistry, Physics, Zoology, Botany Microbiology, Biotechnology, Electronics) OR B.Tech (Biomedical, Electrical & Electronics, Medical Electronics, Electronics & communication, Applied Electronics & Instrumentation)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	06/02/2029
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-5.5-LS-05103-2026-V1-LSSSDC
NQR Version	1.0



Qualification Pack

LFS/N6401: Introduction to IVD, Biomedical Instrumentations and Physiological Measurements

Description

This NOS covers the foundational understanding and practical application of biomedical instrumentation principles, bioelectric signal measurement, electrode theory, and transducer technology used in In-Vitro Diagnostics (IVD) and physiological monitoring.

Scope

The scope covers the following :

- Basic of Biomedical Instrumentation
- Physiological Measurements and IVD

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Accurately interpret and measure physiological signals.

PC2. Operate and calibrate biomedical transducers.

PC3. Implement safe man-machine interfacing and signal conditioning circuits.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Biomedical instrumentation fundamentals, origin of action/resting potentials, electrode theory, transducer classification, signal conditioning.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ethical responsibility in handling human/animal data. Safety-conscious approach. Teamwork and reporting skills.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Accurately interpret and measure physiological signals.	5	5	-	-
PC2. Operate and calibrate biomedical transducers.	5	5	-	-
PC3. Implement safe man-machine interfacing and signal conditioning circuits.	5	5	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6401
NOS Name	Introduction to IVD, Biomedical Instrumentations and Physiological Measurements
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.25
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6402: Bio-Signal Acquisition and Safety in Biomedical Instrumentation

Description

This NOS covers the skills required to accurately acquire, monitor, and record bio-signals from medical diagnostic and monitoring equipment in clinical environments.

Scope

The scope covers the following :

- Bio-Signal Acquisition and Safety in Biomedical Instrumentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Perform bio-signal acquisition in clinical or laboratory settings.
- PC2.** ensure accurate data collection.
- PC3.** monitor bio-signals using biomedical instruments while ensuring patient safety, signal integrity, and compliance with standard operating procedures and regulatory norms.
- PC4.** record bio-signals using biomedical instruments while ensuring patient safety, signal integrity, and compliance with standard operating procedures and regulatory norms.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Familiarity with medical equipment standards
- KU2.** Basic understanding of regulatory norms in clinical environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate attention to detail, teamwork, ethical conduct, and effective communication to ensure safe and accurate diagnostic outcomes.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Perform bio-signal acquisition in clinical or laboratory settings.	4	4	-	-
PC2. ensure accurate data collection.	3	3	-	-
PC3. monitor bio-signals using biomedical instruments while ensuring patient safety, signal integrity, and compliance with standard operating procedures and regulatory norms.	4	4	-	-
PC4. record bio-signals using biomedical instruments while ensuring patient safety, signal integrity, and compliance with standard operating procedures and regulatory norms.	4	4	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6402
NOS Name	Bio-Signal Acquisition and Safety in Biomedical Instrumentation
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.00
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6403: Analytical Principles & Instruments Used for Clinical Chemistry

Description

This NOS covers the skills required to perform analytical testing in clinical chemistry laboratories using various instrumentation.

Scope

The scope covers the following :

- The scope of this NOS is analytical Principles & Instruments Used for Clinical Chemistry

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Operate, calibrate, and maintain clinical chemistry analyzers as per standard protocols

PC2. apply analytical principles such as spectrophotometry, colorimetry, and electrochemistry in test procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Basic analytical chemistry principles, Beer-Lambert law, photometry, and electrochemistry.

KU2. Working principles of clinical chemistry analyzers and sensors.

KU3. Quality assurance concepts, including precision, accuracy, and linearity.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Attention to detail and analytical thinking.

GS2. Effective communication with laboratory and clinical staff.

GS3. Problem-solving and decision-making during technical issues.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Operate, calibrate, and maintain clinical chemistry analyzers as per standard protocols	7	7	-	-
PC2. apply analytical principles such as spectrophotometry, colorimetry, and electrochemistry in test procedures.	8	8	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6403
NOS Name	Analytical Principles & Instruments Used for Clinical Chemistry
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.5
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6404: Operations of Biomedical Transducers and Sensors in IVD

Description

This NOS covers the skills required to operate, test, calibrate, and maintain biomedical transducers and sensors used in In-Vitro Diagnostic (IVD) systems.

Scope

The scope covers the following :

- The scope of this NOS is Operations of Biomedical Transducers and Sensors in IVD

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** accurately operate and maintain various biomedical sensors and transducers used in IVD systems.
- PC2.** ensure reliable diagnostic data, equipment efficiency, and compliance with biomedical safety and quality protocols.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic principles of transduction and signal conversion (pressure, temperature, optical, biochemical).
- KU2.** Working mechanisms of IVD sensors and their clinical applications.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Attention to detail and precision in handling instruments.
- GS2.** Problem-solving and analytical thinking in diagnosing equipment issues.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. accurately operate and maintain various biomedical sensors and transducers used in IVD systems.	7	7	-	-
PC2. ensure reliable diagnostic data, equipment efficiency, and compliance with biomedical safety and quality protocols.	8	8	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6404
NOS Name	Operations of Biomedical Transducers and Sensors in IVD
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.00
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6405: Internet of Things and Smart Systems for IVDs and Medical Devices

Description

This NOS covers on developing the ability to integrate, configure, and maintain IoT-enabled smart systems used in In-Vitro Diagnostic (IVD) and medical devices.

Scope

The scope covers the following :

- The scope of this NOS is IoT-enabled smart systems in IVD and medical devices

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** effectively manage IoT-enabled IVD systems
- PC2.** ensure reliable data communication, enhanced device efficiency and proactive maintenance.
- PC3.** comply with digital health and regulatory standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Familiarity with biomedical device components, sensor types, and data acquisition processes.
- KU2.** Knowledge of data analytics, AI applications, and predictive maintenance in healthcare.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Attention to detail and precision in handling instruments.
- GS2.** Problem-solving and analytical thinking in diagnosing equipment issues.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. effectively manage IoT-enabled IVD systems	5	5	-	-
PC2. ensure reliable data communication, enhanced device efficiency and proactive maintenance.	5	5	-	-
PC3. comply with digital health and regulatory standards.	5	5	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6405
NOS Name	Internet of Things and Smart Systems for IVDs and Medical Devices
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	0.75
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6406: Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics

Description

This NOS covers the skills required to perform application testing, validation, and verification of In-Vitro Diagnostic (IVD) systems used in hematology, immunoassay, and molecular diagnostics.

Scope

The scope covers the following :

- The Scope of this NOS is Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Operate hematology, immunoassay, and molecular diagnostic analyzers.
- PC2.** validate and verify the performance of IVD systems
- PC3.** ensure consistent, accurate, and compliant diagnostic testing across hematology, immunoassay, and molecular platforms

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of calibration, validation, and verification protocols for IVD systems.
- KU2.** Familiarity with troubleshooting methodologies and root cause analysis in testing processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Time management and adherence to testing schedules.
- GS2.** Professional ethics and accountability in maintaining test integrity and data confidentiality.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Operate hematology, immunoassay, and molecular diagnostic analyzers.	5	5	-	-
PC2. validate and verify the performance of IVD systems	5	5	-	-
PC3. ensure consistent, accurate, and compliant diagnostic testing across hematology, immunoassay, and molecular platforms	5	5	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6406
NOS Name	Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.75
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6407: Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs

Description

This NOS covers the skills required to implement design control and risk management processes during the development, validation, and lifecycle management of In-Vitro Diagnostic (IVD) medical devices.

Scope

The scope covers the following :

- The scope of this NOS is Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Manage design and risk processes effectively

PC2. Ensure that IVD medical devices meet global regulatory requirements

PC3. maintain safety and performance standards, and align with clinical and quality compliance frameworks.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Familiarity with clinical evaluation processes and post-market surveillance requirements.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Time management and adherence to testing schedules.

GS2. Professional ethics and accountability in maintaining test integrity and data confidentiality.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Manage design and risk processes effectively	5	5	-	-
PC2. Ensure that IVD medical devices meet global regulatory requirements	5	5	-	-
PC3. maintain safety and performance standards, and align with clinical and quality compliance frameworks.	5	5	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6407
NOS Name	Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	3.00
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6408: Operations and Servicing of Point of Care IVDs

Description

This NOS cover the skills required to operate, calibrate, and service Point-of-Care (POC) IVD devices to ensure accurate diagnostic testing, optimal performance, and adherence to safety and regulatory quality standards.

Scope

The scope covers the following :

- The Scope of this NOS is Operations and Servicing of Point of Care IVDs

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install, operate, and calibrate Point-of-Care (POC) IVD analyzers.
- PC2.** manage the operation and servicing of POC IVD equipment
- PC3.** ensure reliable performance, minimal downtime, and accurate diagnostic outcomes in compliance with industry and quality standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of calibration, validation, and maintenance protocols.
- KU2.** Familiarity with biomedical instrumentation, sensor technology, and reagent handling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Customer orientation and professionalism in handling field operations.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	15	-	-
PC1. Install, operate, and calibrate Point-of-Care (POC) IVD analyzers.	5	5	-	-
PC2. manage the operation and servicing of POC IVD equipment	5	5	-	-
PC3. ensure reliable performance, minimal downtime, and accurate diagnostic outcomes in compliance with industry and quality standards.	5	5	-	-
NOS Total	15	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6408
NOS Name	Operations and Servicing of Point of Care IVDs
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	1.25
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

LFS/N6409: Perform installation, routine maintenance and trouble shooting following the guidelines for IVD component assembly and material handling guidelines.

Description

This NOS cover the skills required to perform installation, calibration, preventive maintenance, and troubleshooting of In Vitro Diagnostic (IVD) equipment, ensuring accurate setup, optimal performance, and compliance with regulatory and safety standards.

Scope

The scope covers the following :

- The Scope of this NOS is performing installation, routine maintenance and trouble shooting following the guidelines for IVD component assembly and material handling guidelines

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Perform accurate installation and calibration of IVD equipment as per manufacturer's specifications.
- PC2.** Conduct initial quality checks and safety validations before handover.
- PC3.** Maintain complete installation documentation and customer sign-off reports.
- PC4.** Ensure compliance with relevant medical device standards during installation.
- PC5.** Execute scheduled preventive maintenance as per defined SOPs.
- PC6.** Troubleshoot and repair minor faults independently with proper tools.
- PC7.** Replace faulty components and restore equipment functionality efficiently.
- PC8.** Maintain service logs and suggest improvement actions to reduce repeat failures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of IVD device specifications
- KU2.** Installation protocols and safety guidelines
- KU3.** Understanding of system diagnostics and failure points
- KU4.** Quality assurance protocols

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Attention to detail Planning and organization Analytical thinking Precision and consistency

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	15	-	-
PC1. Perform accurate installation and calibration of IVD equipment as per manufacturer's specifications.	1	1	-	-
PC2. Conduct initial quality checks and safety validations before handover.	1	2	-	-
PC3. Maintain complete installation documentation and customer sign-off reports.	1	2	-	-
PC4. Ensure compliance with relevant medical device standards during installation.	1	2	-	-
PC5. Execute scheduled preventive maintenance as per defined SOPs.	1	2	-	-
PC6. Troubleshoot and repair minor faults independently with proper tools.	1	2	-	-
PC7. Replace faulty components and restore equipment functionality efficiently.	2	2	-	-
PC8. Maintain service logs and suggest improvement actions to reduce repeat failures.	2	2	-	-
NOS Total	10	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6409
NOS Name	Perform installation, routine maintenance and trouble shooting following the guidelines for IVD component assembly and material handling guidelines.
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	6.00
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

LFS/N6410: Manage Customer Satisfaction through field service and maintenance operations

Description

This NOS Covers the skills required to manage on-site field service operations for IVD and medical devices, ensuring timely issue resolution, professional customer interaction, accurate service documentation, and continuous improvement in customer satisfaction levels.

Scope

The scope covers the following :

- The scope of this NOS is about Manage Customer Satisfaction through field service and maintenance operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Handle customer service requests with professionalism and timely response.
- PC2.** Diagnose field-level issues effectively and provide immediate solutions or escalation.
- PC3.** Maintain clear communication with lab staff, healthcare professionals, and other stakeholders.
- PC4.** Log service details accurately and ensure customer satisfaction post-service and maintain the C-SAT Score.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Service protocols and escalation matrix ,Customer complaint handling procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Active listening and empathy Professional communication



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	15	-	-
PC1. Handle customer service requests with professionalism and timely response.	2	3	-	-
PC2. Diagnose field-level issues effectively and provide immediate solutions or escalation.	2	4	-	-
PC3. Maintain clear communication with lab staff, healthcare professionals, and other stakeholders.	3	4	-	-
PC4. Log service details accurately and ensure customer satisfaction post-service and maintain the C-SAT Score.	3	4	-	-
NOS Total	10	15	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6410
NOS Name	Manage Customer Satisfaction through field service and maintenance operations
Sector	Life Sciences
Sub-Sector	
Occupation	POST SALES SERVICES -MEDICAL DEVICES AND IVD
NSQF Level	5.5
Credits	4.5
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by Life Sciences Sector Skill Development Council (LSSSDC)
2. Each Element will be assigned marks proportional to its importance in NOS. LSSSDC will also lay down the proportion of marks for Theory, Practical, Project, and Viva for each Element.
3. The assessment for the theory part will be based on the knowledge bank of questions created by the LSSSDC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. LSSSDC as an assessment and awarding body will create unique evaluations for each assessment component i.e. theory, practical, project, and viva for every student at each examination/training center based on this criterion.
6. Wherever any assessment component is not applicable/ feasible, the balance assessment components will be used to assess the candidate, and accordingly the total marks will be calculated only for the applied



Qualification Pack

assessment component.

7. To pass the Qualification Pack, every trainee should score a minimum of 50%-70% of marks in each NOS (depending on NSQF Level) to successfully clear the assessment. In the case of a Govt funded program, the program guidelines will be overarching on the pass percentage rules.

8. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6401.Introduction to IVD, Biomedical Instrumentations and Physiological Measurements	15	15	0	0	30	10
LFS/N6402.Bio-Signal Acquisition and Safety in Biomedical Instrumentation	15	15	0	0	30	10
LFS/N6403.Analytical Principles & Instruments Used for Clinical Chemistry	15	15	0	0	30	10
LFS/N6404.Operations of Biomedical Transducers and Sensors in IVD	15	15	0	0	30	10
LFS/N6405.Internet of Things and Smart Systems for IVDs and Medical Devices	15	15	0	0	30	5

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6406.Application Testing of IVDs for Hematology, Immunoassay and Molecular Diagnostics	15	15	0	0	30	10
LFS/N6407.Medical Device Design Control and Risk Management as per Regulatory Quality Standards and Clinical guidelines for IVDs	15	15	0	0	30	10
LFS/N6408.Operations and Servicing of Point of Care IVDs	15	15	0	0	30	10
LFS/N6409.Perform installation, routine maintenance and trouble shooting following the guidelines for IVD component assembly and material handling guidelines.	10	15	0	0	25	10
LFS/N6410.Manage Customer Satisfaction through field service and maintenance operations	10	15	0	0	25	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	160	180	-	-	340	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.